NutriPath

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Guide
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Inroduction

- Hard to plan personalized meals
- Generic plans ignore health needs
- AI can give unsafe or inaccurate meals
- NutriPath takes the user’s health profile
- AI generates personalized meals
- Safety rules validate every meal
- Safer, explainable meal plans

Problem Statement

Current nutrition apps cannot guarantee safe, accurate meal plans for people with medical conditions, so there is a need for a deterministic, safety-validated personalized meal planning system.

Project Overview



Problem

Generic and AI-generated meal plans may ignore medical conditions, allergies, dietary preferences, and nutritional limits, resulting in unsuitable recommendations.



Solution

NutriPath generates personalized meal plans and validates them using trusted nutritional data and safety rules before displaying them.



Why It Matters

It helps users receive safer, more suitable, and explainable nutrition recommendations.

Literature Survey

Sl. No.	Paper / Framework	Methodology	Key Merits	Limitation / Gap
1	NutriOrion (arXiv:2602.18650, 2026)	Hard drug-nutrient constraints; ADIME audit trail	Avoids the “Healthy Food Trap” (e.g., Warfarin and Vitamin K)	Server-side only; no offline mobile client
2	NutriGen (arXiv:2502.20601, 2025)	Deterministic nutrient math, separate from recipe generation	Exact nutrient values; no numerical hallucination	Limited adaptation when a user rejects an ingredient
3	HealthGenie (arXiv:2504.14594, 2025)	Interactive ingredient exclusion with clinical substitution	Real-time customization with clinical compliance	Needs local storage and cloud sync

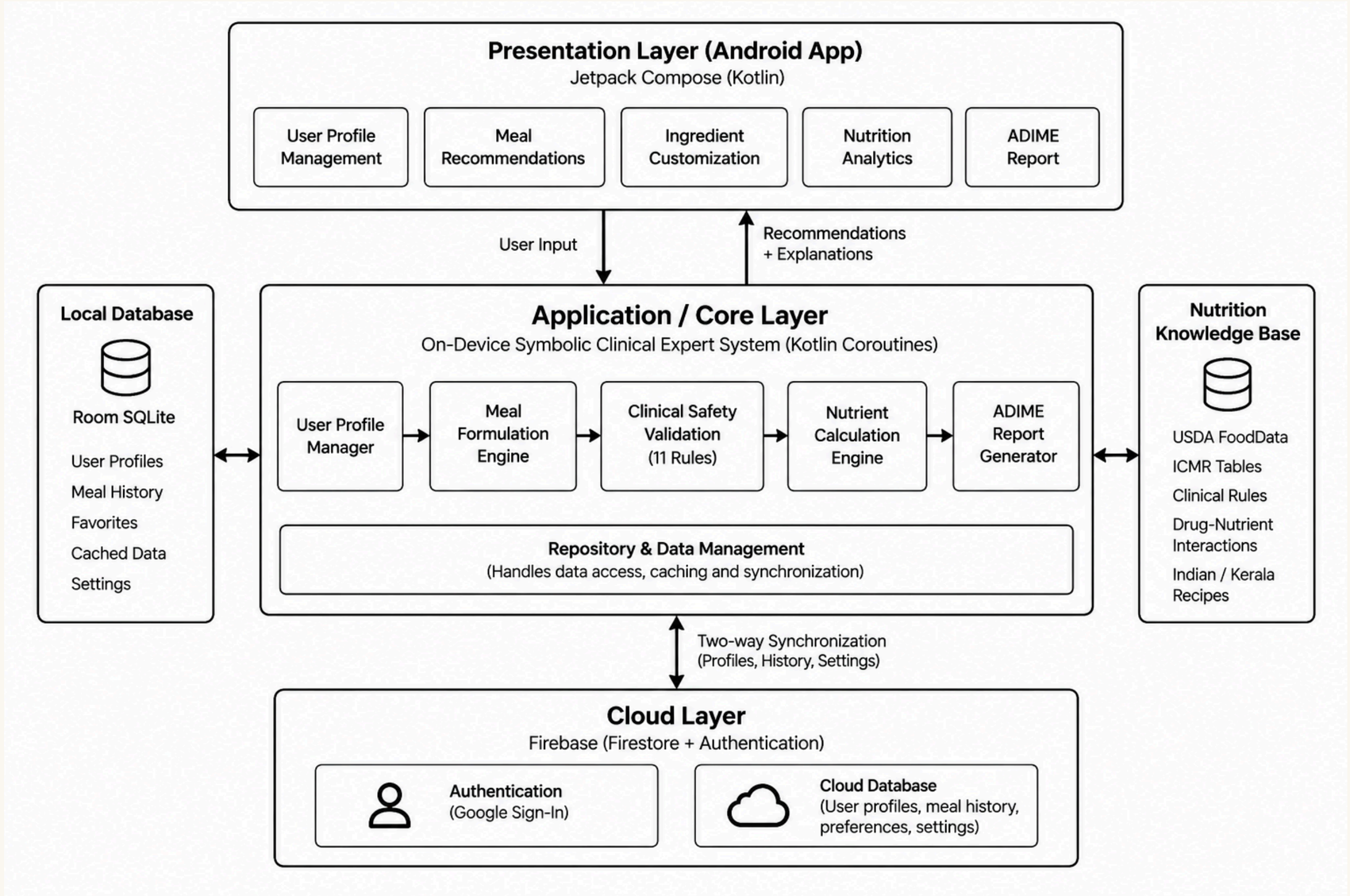
Gap Identified

- Existing apps skip complete safety validation
- Generative AI nutrient values are unreliable
- Few combine allergies, conditions, medications, and nutrient limits
- Most need internet and respond slowly
- No clinical audit notes (ADIME)
- Rigid plans when an ingredient is excluded
- NutriPath: on-device, rule-based, safety-validated, offline-first

Proposed Solution

- Collect health profile and goals
- Calculate calorie and nutrient targets
- Generate candidate meals with rules
- Compute exact nutrients from USDA / ICMR data
- Validate with 11 safety rules
- Reject or modify unsafe meals
- Show Approved / Modified / Warning / Rejected badges
- Explain every decision with an ADIME record

System Architecture



Architecture Overview

NutriPath uses Jetpack Compose for the UI and Kotlin for application logic. Its on-device clinical expert system combines meal formulation, deterministic nutrient calculation, 11-rule safety validation, ingredient substitution, and ADIME report generation. Room SQLite provides offline storage, USDA/ICMR tables supply nutrition data, and Firebase Authentication and Firestore enable secure sign-in and cloud synchronization.

NUTRIPATH

Architecture Components

1. Client Layer

- Provides the user interface using Jetpack Compose (Home, Meal Recommendations, Nutrition Analytics, and ADIME Reports).
- Collects user profiles, medical conditions, medications, and allergies.
- Displays personalized meal plans, nutrient information, and clinical safety results.

2. Application / Core Layer

- Meal Formulation Engine: Generates personalized meal plans.
- Clinical Safety Validator: Applies 11 rule-based checks for medical conditions, allergies, and drug–nutrient interactions.
- Nutrient Calculation Engine: Calculates nutrients using USDA/ICMR food composition data.
- Adaptation & Substitution Engine: Replaces excluded ingredients with suitable alternatives.
- ADIME Generator: Produces clinical audit reports.

3. Local Storage Layer

- Room SQLite: Stores user profiles, meal plans, meal history, and cached food data.
- Supports offline access to core nutrition and safety features.
- Repository & Data Management: Coordinates data access, caching, and synchronization.

4. Cloud Layer

- Firebase Authentication: Supports Google Sign-In.
- Cloud Firestore: Stores and synchronizes user profiles, meal plans, and preferences.
- Enables two-way synchronization and access across multiple devices.

MODULE DESCRIPTION

1. User Authentication & Profile Module

- Supports Google Sign-In and email/password authentication.
- Collects user details, medical conditions, medications, and allergies.

2. Meal Recommendation Module

- Generates personalized daily meal plans based on the user's health profile.
- Supports breakfast, lunch, dinner, and snacks.

3. Nutrition Calculation Module

- Calculates calories and nutrients using USDA and ICMR food composition tables.
- Provides deterministic nutritional values based on ingredient quantities.

4. Clinical Safety Validation Module

- Applies 11 rule-based safety checks for medical conditions, allergies, and drug–nutrient interactions.
- Classifies meals as Approved, Modified, Warning, or Rejected.

MODULE DESCRIPTION

5. Ingredient Adaptation Module

- Allows users to exclude unwanted ingredients.
- Suggests suitable alternatives and recalculates nutritional values.

6. ADIME Report Module

- Generates clinical reports covering Assessment, Diagnosis, Intervention, Monitoring, and Evaluation.
- Provides explanations for meal recommendations and safety decisions.

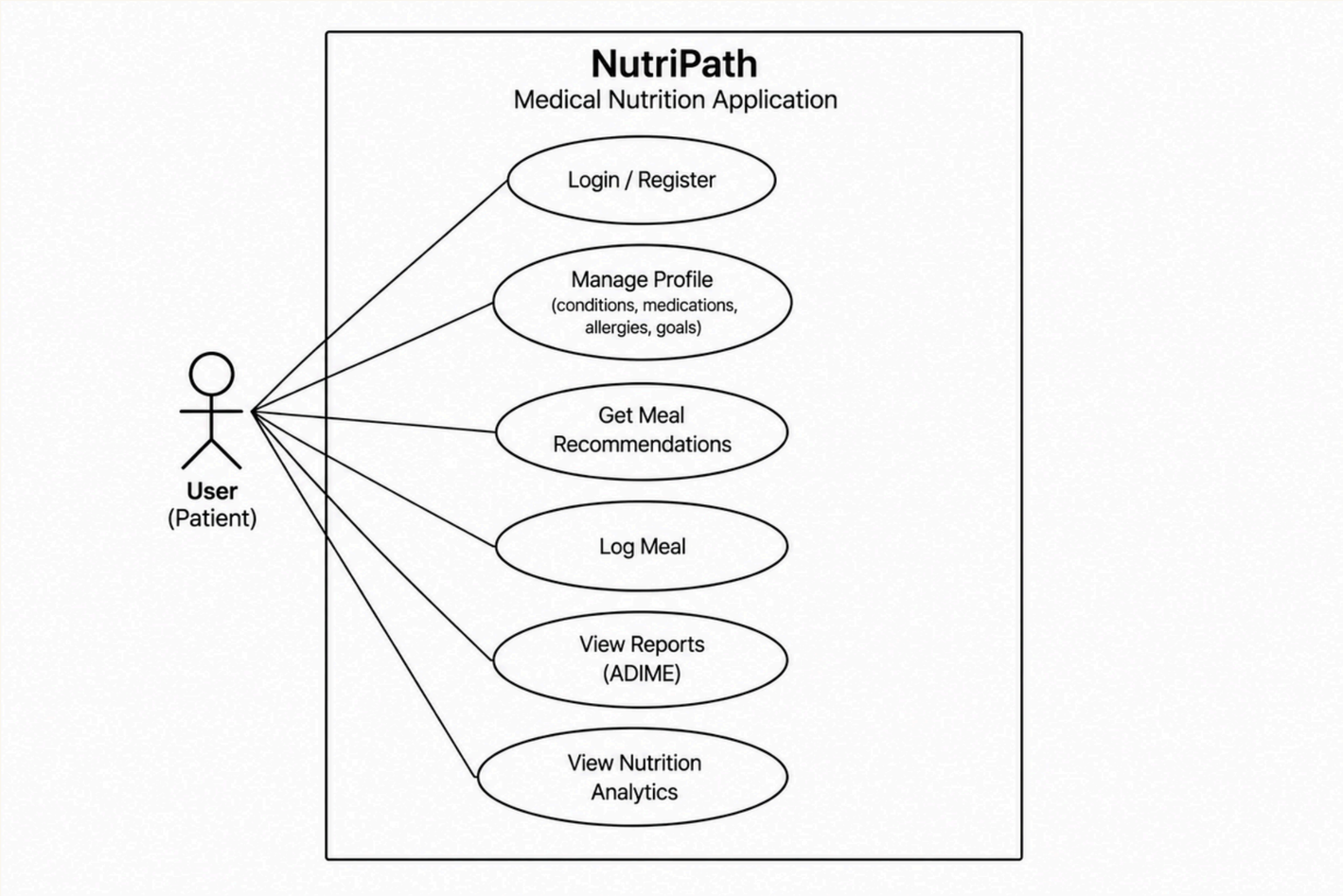
7. Daily Nutrition Tracking Module

- Tracks planned and consumed meals separately.
- Updates daily calorie and nutrient totals when meals are logged.

8. Data Storage & Synchronization Module

- Uses Room SQLite for local storage and offline access.
- Uses Firebase Firestore for cloud storage and synchronization across devices.

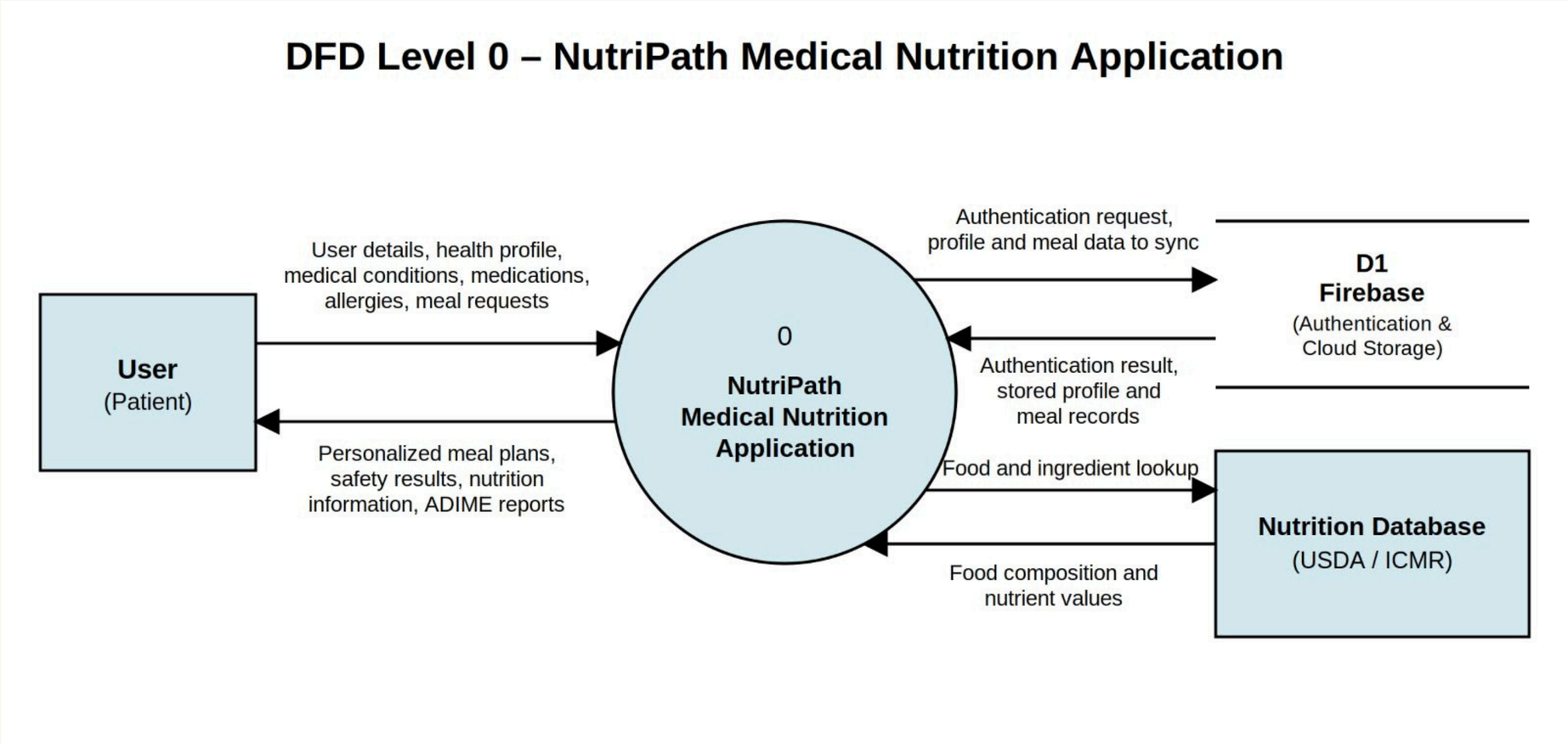
USE CASE DIAGRAM



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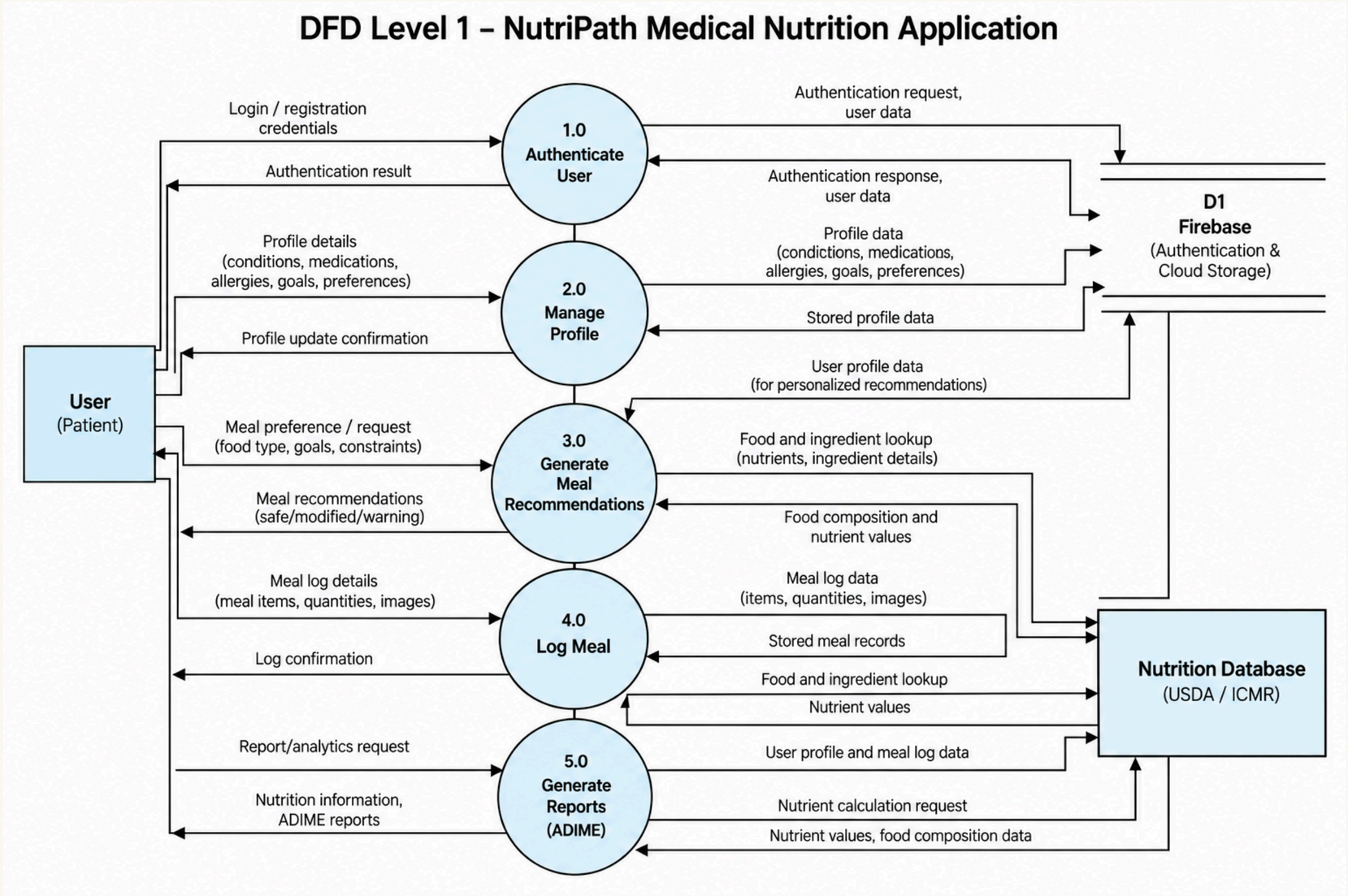
DATA FLOW DIAGRAM

Level 0



NUTRIPATH

Level 1



SOFTWARE REQUIREMENT SPECIFICATION

User Interface

- The system shall provide a simple and responsive user interface using Jetpack Compose.
- Users shall register and log in securely using Firebase Authentication, including Google Sign-In.
- Users shall enter and manage their medical conditions, medications, allergies, dietary preferences, and nutritional goals.
- Users shall view personalized meal recommendations with nutrient information and clinical safety validation statuses.
- Users shall log consumed meals, track daily nutritional intake, and view nutrition analytics.
- Users shall generate and view ADIME nutrition reports based on user profiles and meal records.
- The system shall provide input validation, clear safety warnings, and informative feedback for user actions.

SOFTWARE REQUIREMENT SPECIFICATION

- Operating System: Windows / Linux / macOS.
- Frontend: Kotlin and Jetpack Compose for the Android user interface.
- Programming Language: Kotlin.
- Application Architecture: MVVM with Repository pattern.
- Clinical Logic: On-device deterministic clinical expert system with 11 rule-based safety checks.
- Nutrition Calculation: Nutrient calculation engine using USDA and ICMR food composition data.
- Local Database: Room with SQLite for offline data storage.
- Cloud Services: Firebase Authentication for secure login and Google Sign-In; Cloud Firestore for user profiles, meal plans, and data synchronization.
- Data Management: Repository layer for data access, caching, and synchronization between local storage and Firebase.
- Reporting: ADIME nutrition report generation.
- Development Tool: Android Studio.
- Build System: Gradle.
- Target Platform: Android devices.

HARDWARE REQUIREMENTS

- Processor: Intel Core i3 or equivalent and above.
- RAM: Minimum 4 GB; 8 GB recommended.
- Storage: Minimum 10 GB of free disk space for development tools and project files.
- Mobile Device: Android smartphone running Android 8.0 (Oreo) or above.
- Display: Minimum 1366 × 768 resolution for development.
- Internet Connection: Required for Firebase Authentication, Google Sign-In, cloud synchronization, and initial data downloads.
- Graphics: Integrated graphics sufficient for application development.
- Development Device: A computer capable of running Android Studio and the Android Emulator.

FUNCTIONAL REQUIREMENTS

- User Authentication: Users shall be able to register, log in, log out, and securely access protected features using Firebase Authentication.
- Health Profile Management: Users shall be able to enter and update their medical conditions, medications, allergies, dietary preferences, and nutritional goals.
- Personalized Meal Generation: The system shall generate personalized meal recommendations based on the user's health profile and nutritional requirements.
- Nutrition Calculation: The system shall calculate calories and nutrient values using trusted USDA and ICMR food composition data.
- Clinical Safety Validation: The system shall validate meals against allergens, medical restrictions, dietary requirements, and nutrient limits using an on-device rule-based clinical expert system.
- Meal Classification: The system shall classify meal recommendations as Approved, Modified, Warning, or Rejected based on safety validation results.
- Meal Plan Management: Users shall be able to view, save, and access previously generated meal plans.
- Safety Explanations: The system shall display explanations for meal approval, modification, warnings, or rejection.
- Ingredient Adaptation: The system shall identify unsuitable ingredients and suggest appropriate substitutions when possible.
- Daily Nutrition Tracking: Users shall be able to log consumed meals and monitor their daily calorie and nutrient intake.
- Reports and Logs: The system shall store meal plans, meal records, and clinical validation results locally using Room/SQLite and synchronize supported data with Firebase.
- ADIME Report Generation: The system shall generate nutrition reports containing assessment, diagnosis, intervention, and monitoring/evaluation information based on available user and meal data.

DATABASE REQUIREMENTS

- **Local Database**: Room with SQLite shall be used to store user profiles, medical conditions, allergies, meal plans, meal history, and nutrition tracking records.
- **Cloud Database**: Firebase Cloud Firestore shall store and synchronize supported user profiles, preferences, meal plans, and related records across devices.
- **Food Composition Database**: USDA and ICMR food composition data shall provide calorie and nutrient values for food items and ingredients.
- **User Data Management**: The database shall support creating, retrieving, updating, and deleting user-owned records as appropriate.
- **Offline Access:** The local database shall allow users to access saved meal plans and relevant nutrition data without an internet connection.
- **Data Synchronization:** The system shall synchronize supported local and cloud records when connectivity is available.
- **Clinical Safety Rules:** Medical restrictions, allergen information, nutrient limits, and validation rules shall be stored and managed in a structured format suitable for deterministic safety checks.
- **Data Integrity:** The system shall maintain consistent relationships between user profiles, meal plans, ingredients, nutrient values, and validation results.
- **Security and Privacy:** Sensitive health information shall be protected through appropriate access controls and secure data handling.
- **Backup and Recovery**: Cloud-synchronized records shall support recovery when users reinstall the application or switch devices.

NON-FUNCTIONAL REQUIREMENTS

- **Performance**: The system shall generate meal recommendations and perform clinical safety validation within a reasonable response time.
- **Reliability**: The system shall provide consistent nutrition calculations and deterministic safety validation results for the same inputs and rules.
- **Security**: The system shall protect user accounts and sensitive health information through secure authentication and appropriate access controls.
- **Usability**: The application shall provide a simple, intuitive, and user-friendly interface for entering health information and viewing meal recommendations.
- **Availability**: Core meal recommendation and safety validation features shall remain available offline when the required data is stored locally.
- **Scalability**: The cloud infrastructure shall support increasing numbers of users, profiles, and meal records.
- **Maintainability**: The modular Kotlin architecture shall allow individual components, such as nutrition calculation and safety validation, to be updated independently.
- **Data Integrity**: The system shall maintain accurate and consistent user profiles, food composition data, meal plans, and validation results.
- **Compatibility**: The application shall support compatible Android devices and screen sizes.
- **Privacy**: The system shall collect and store only necessary user information and restrict access to authorized users.
- **Interoperability**: The system shall integrate with Firebase services and use USDA/ICMR food composition data in supported formats.
- **Safety**: The system shall validate meal recommendations against configured medical restrictions, allergies, and nutritional limits before displaying their final safety status.

RESULTS

NutriPath Clinical Nutrition



Nutrition is your superpower for change

Scientifically tailored recipes and safety rules built around your body and health conditions.

GET STARTED

I already have an account • Sign In

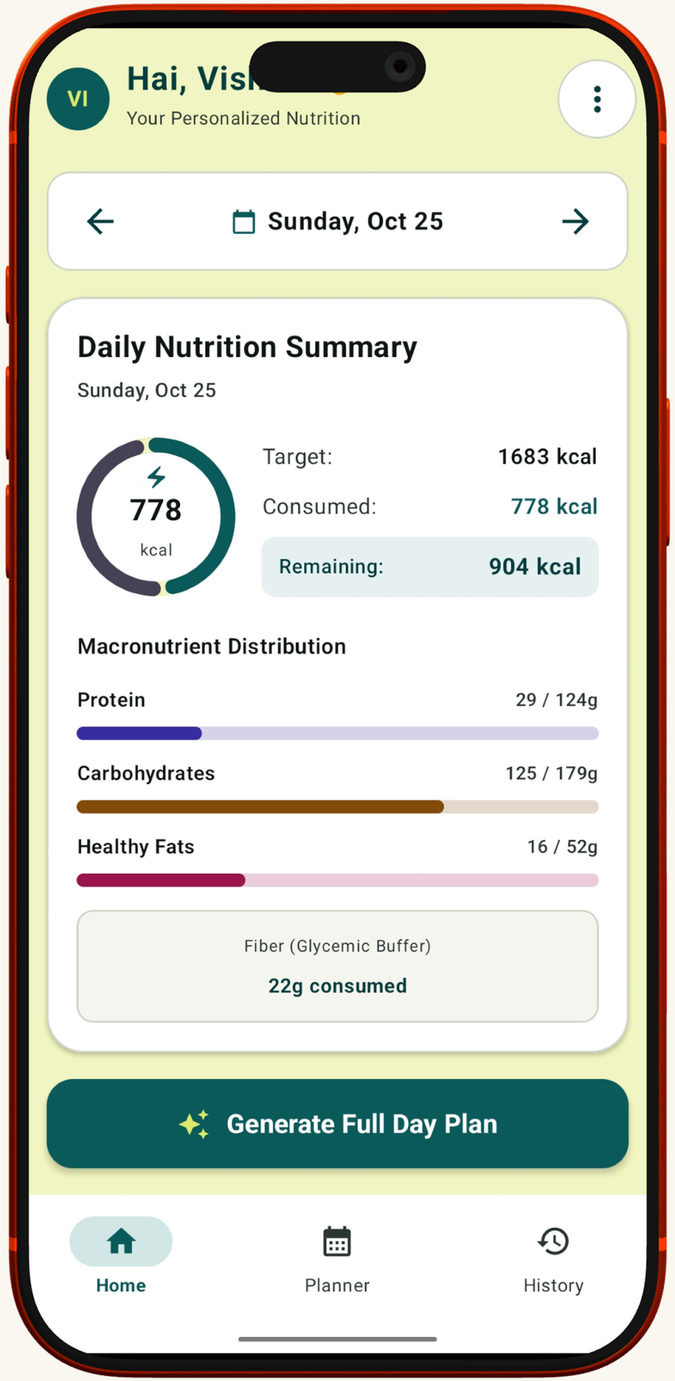
Medical conditions & allergies

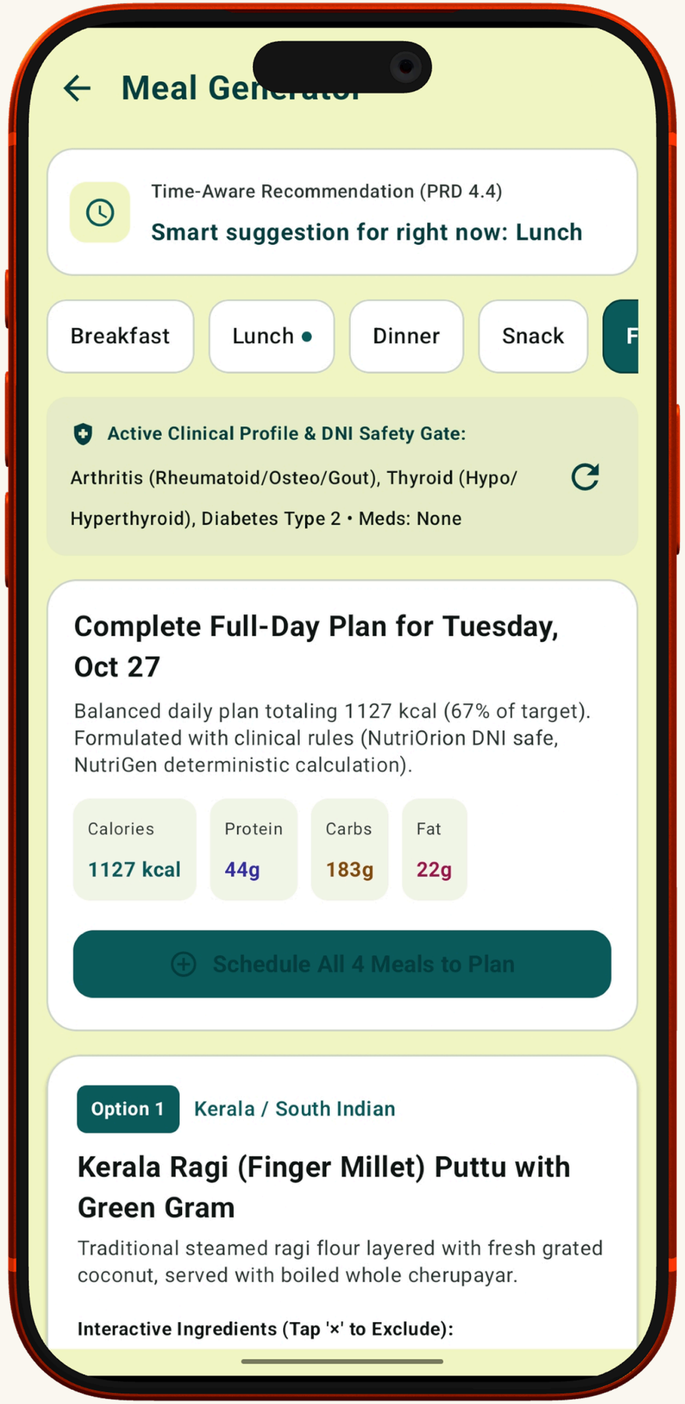
NutriPath uses these to enforce clinical safety limits (DASH sodium, ADA glycemic index, KDIGO renal thresholds) and filter allergens.

Medical Conditions

- [x] Diabetes Type 2
- [ ] Hypertension
- [x] Thyroid (Hypo/Hyperthyroid)
- [x] Arthritis (Rheumatoid/Osteo/Gout)
- [ ] Chronic Kidney Disease (CKD)
- [ ] High Cholesterol / Hyperlipidemia
- [ ] Celiac Disease
- [ ] None / General Wellness

Food Allergies (Zero-tolerance exclusions)


VI
Hai, Vis
Your Personalized Nutrition
Sunday, Oct 25
Daily Nutrition Summary
Sunday, Oct 25
778
kcal
Target: 1683 kcal
Consumed: 778 kcal
Remaining: 904 kcal
Macronutrient Distribution
Protein 29 / 124g
Carbohydrates 125 / 179g
Healthy Fats 16 / 52g
Fiber (Glycemic Buffer)
22g consumed
Generate Full Day Plan
Home
Planner
History


Meal Generator
Time-Aware Recommendation (PRD 4.4)
Smart suggestion for right now: Lunch
Breakfast
Lunch
Dinner
Snack
Active Clinical Profile & DNI Safety Gate:
Arthritis (Rheumatoid/Osteo/Gout), Thyroid (Hypo/ Hyperthyroid), Diabetes Type 2 • Meds: None
Complete Full-Day Plan for Tuesday, Oct 27
Balanced daily plan totaling 1127 kcal (67% of target). Formulated with clinical rules (NutriOrion DNI safe, NutriGen deterministic calculation).
Calories 1127 kcal
Protein 44g
Carbs 183g
Fat 22g
Schedule All 4 Meals to Plan
Option 1
Kerala / South Indian
Kerala Ragi (Finger Millet) Puttu with Green Gram
Traditional steamed ragi flour layered with fresh grated coconut, served with boiled whole cherupayar.
Interactive Ingredients (Tap '×' to Exclude):

CONCLUSION

- NutriPath provides personalized meal recommendations based on users’ health conditions, allergies, and nutritional needs. Using Kotlin, Room, Firebase, and rule-based safety validation, it supports safer food choices, nutrition tracking, and ADIME reporting while complementing professional medical advice.

REFERENCES

Gao, F., Zhao, X., Xia, D., et al. “HealthGenie: Empowering Users with Healthy Dietary Guidance through Knowledge Graph and Large Language Models,” 2025.2504.14594v1.pdf

Khamesian, S., Arefeen, A., Carpenter, S. M., and Ghasemzadeh, H. “NutriGen: Personalized Meal Plan Generator Leveraging Large Language Models to Enhance Dietary and Nutritional Adherence,” 2025.2502.20601v2.pdf

Wu, J., Yan, R., Luo, H., et al. “NutriOrion: A Hierarchical Multi-Agent Framework for Personalized Nutrition Intervention Grounded in Clinical Guidelines,” 2026.2602.18650v1.pdf

World Health Organization. “Sodium Reduction.” Available at: WHO

American Diabetes Association. Standards of Care in Diabetes, 2025.

Indian Council of Medical Research. Dietary Guidelines for Indians, 2024.

The End